

April 15, 2026

SCynergy 2026



Operational GeoAI for Scalable Satellite Analytics

: TerraMind for Multimodal Flood Inference

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TerraMind for Multimodal Flood Mapping on MeluXina

Connecting geospatial data engineering, Earth observation AI, and GPU-based HPC execution — in one hands-on workshop.



Concepts

20 min introduction



Hands-On

60 min guided notebooks



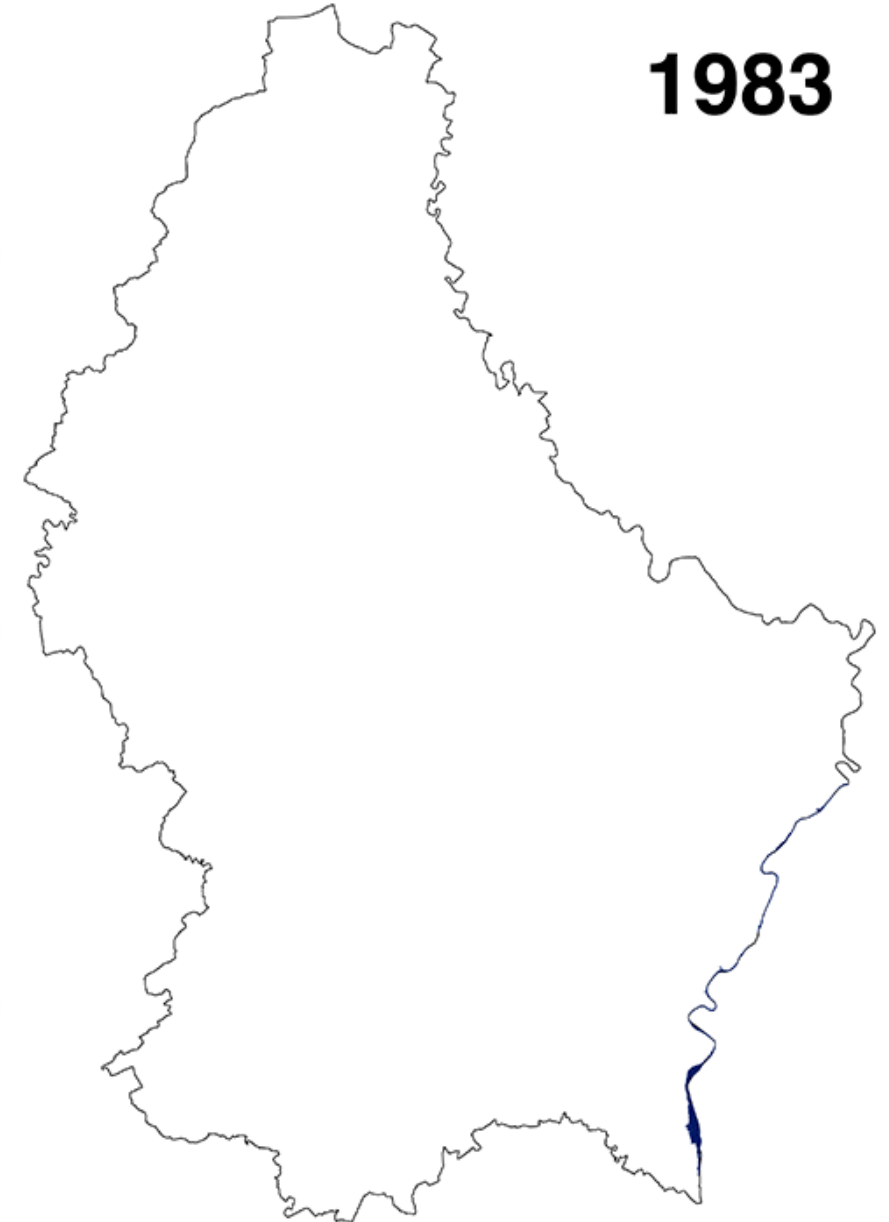
Platform

MeluXina GPU + Open OnDemand JupyterLab



No need to be an expert in all three domains (AI, HPC, Earth AI) — that's what the next 20 minutes are for.

1983



Luxembourg flooding history

Why Floods Need Multimodal GeoAI

Floods vary in location, timing, and detection, so combining multiple data sources is essential for a complete view.

Sentinel-2 (Optical)

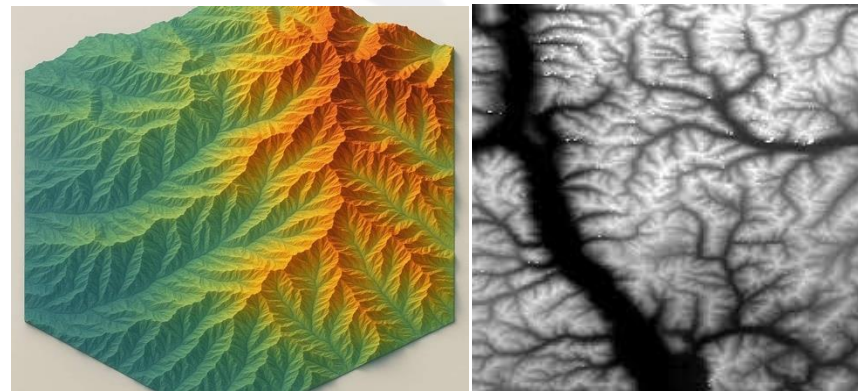
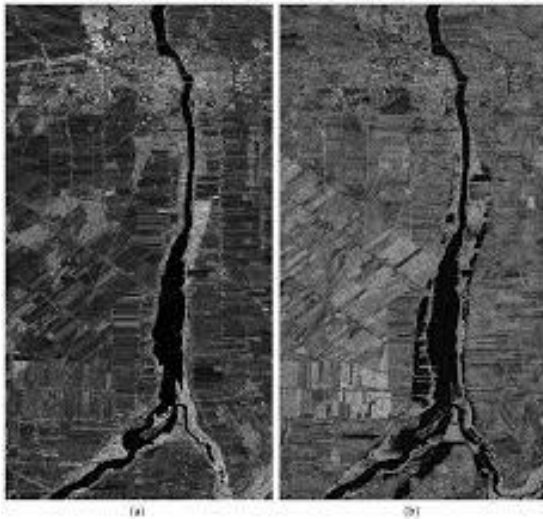
Spectral detail, intuitive RGB — but **blocked by clouds** during active flood events.

Sentinel-1 (SAR)

All-weather radar. Water surfaces appear **dark and distinct** — but harder to interpret visually.

DEM (Elevation)

Topographic context. Static, but **physically meaningful** for flood plausibility and drainage.



 **Key idea:** Flood mapping is not a single-image problem — it is a multimodal spatiotemporal reasoning problem.

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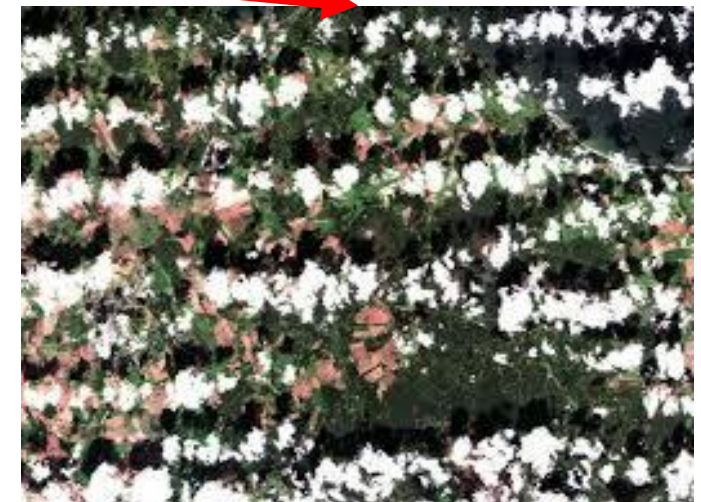
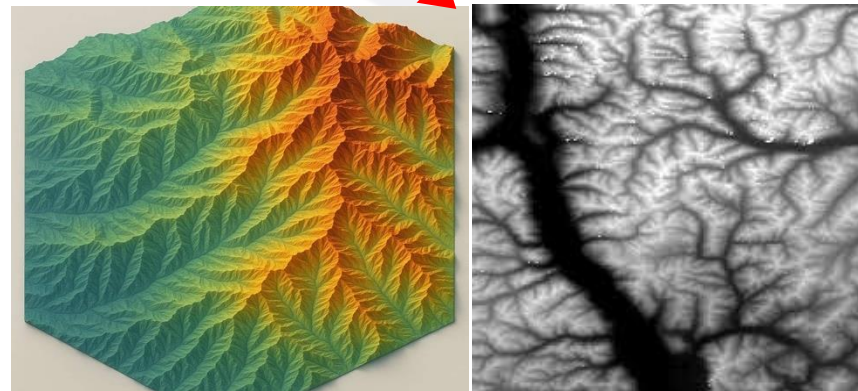
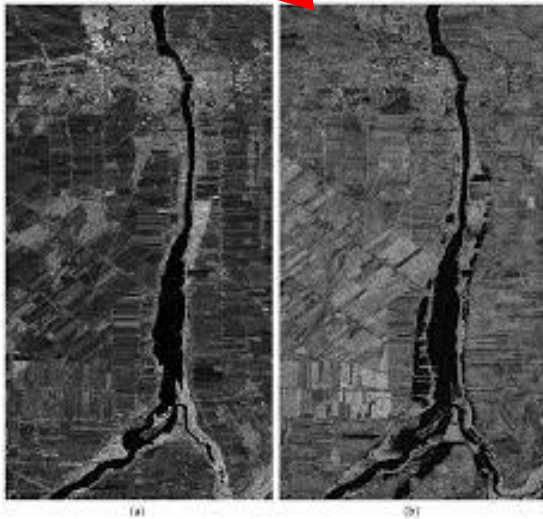
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From GeoAI to Geospatial Foundation Models

What is GeoAI?

AI methods applied to **geospatial data** — satellite imagery, elevation models, vector maps, and more.

What is a GeoFM?

A **large pretrained model** that learns general representations from massive Earth observation datasets, then adapts to downstream tasks — without training from scratch.

GeoFMs offer **reusable representations** and transfer across datasets and tasks — reducing task-specific engineering.

Segmentation

Classification

Change Detection

Retrieval

Generation



Why TerraMind Is a Strong Fit for Multimodal EO

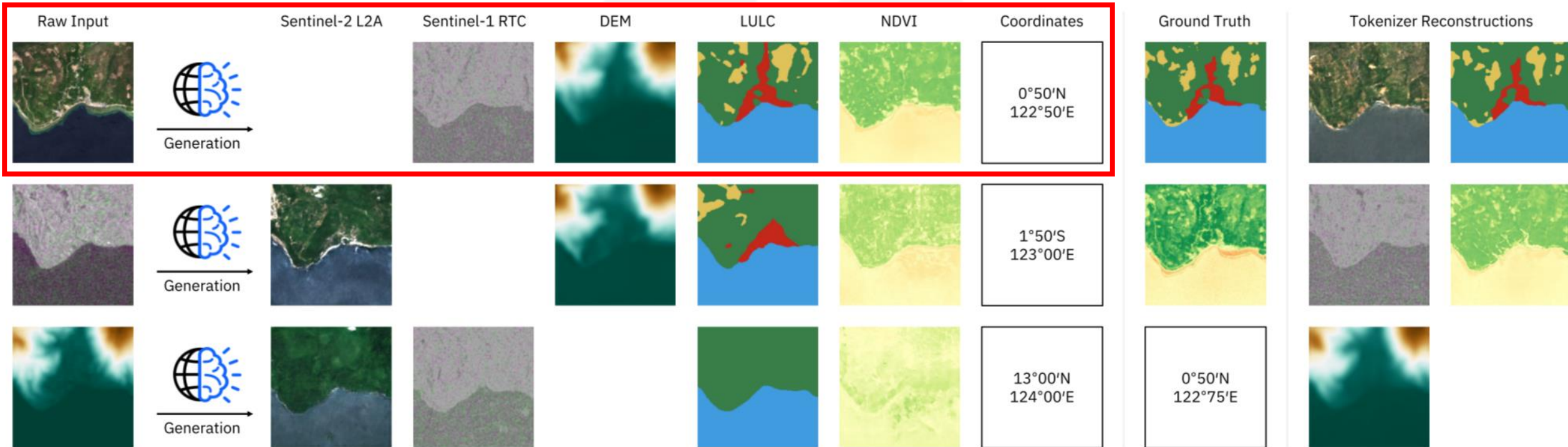
A Different Kind of Foundation Model

TerraMind is a **multimodal generative EO foundation model** — designed for cross-modal reasoning, not just single-image encoding.

- Standard fine-tuning on downstream tasks
- Multimodal inference across S1, S2, and DEM
- Thinking-in-Modalities (TiM) for cross-modal generation

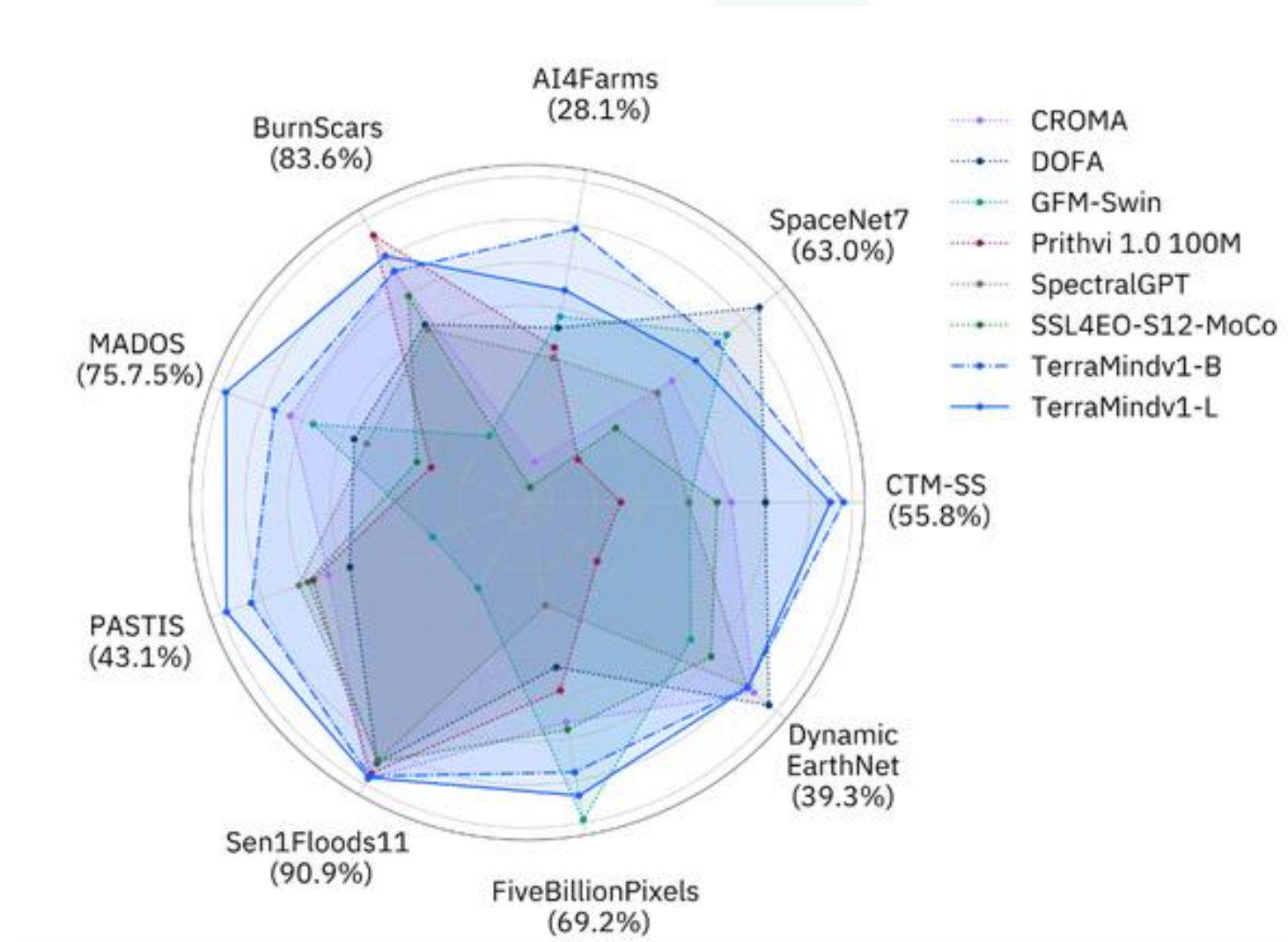
Model Sizes

Tiny/Small → local dev and demos
Base → practical default
Large → best when compute allows





Why TerraMind Is a Strong Fit for Multimodal EO



TerraMind and Prithvi-EO-v2: Different Emphasis

Both are geospatial foundation models for Earth observation — but they're built around different design priorities.

Prithvi-EO-v2

- Spatiotemporal transformer architecture
- **Masked autoencoding (MAE) pretraining**
- Optional time and location encodings
- Strong EO encoder for downstream tasks

TerraMind

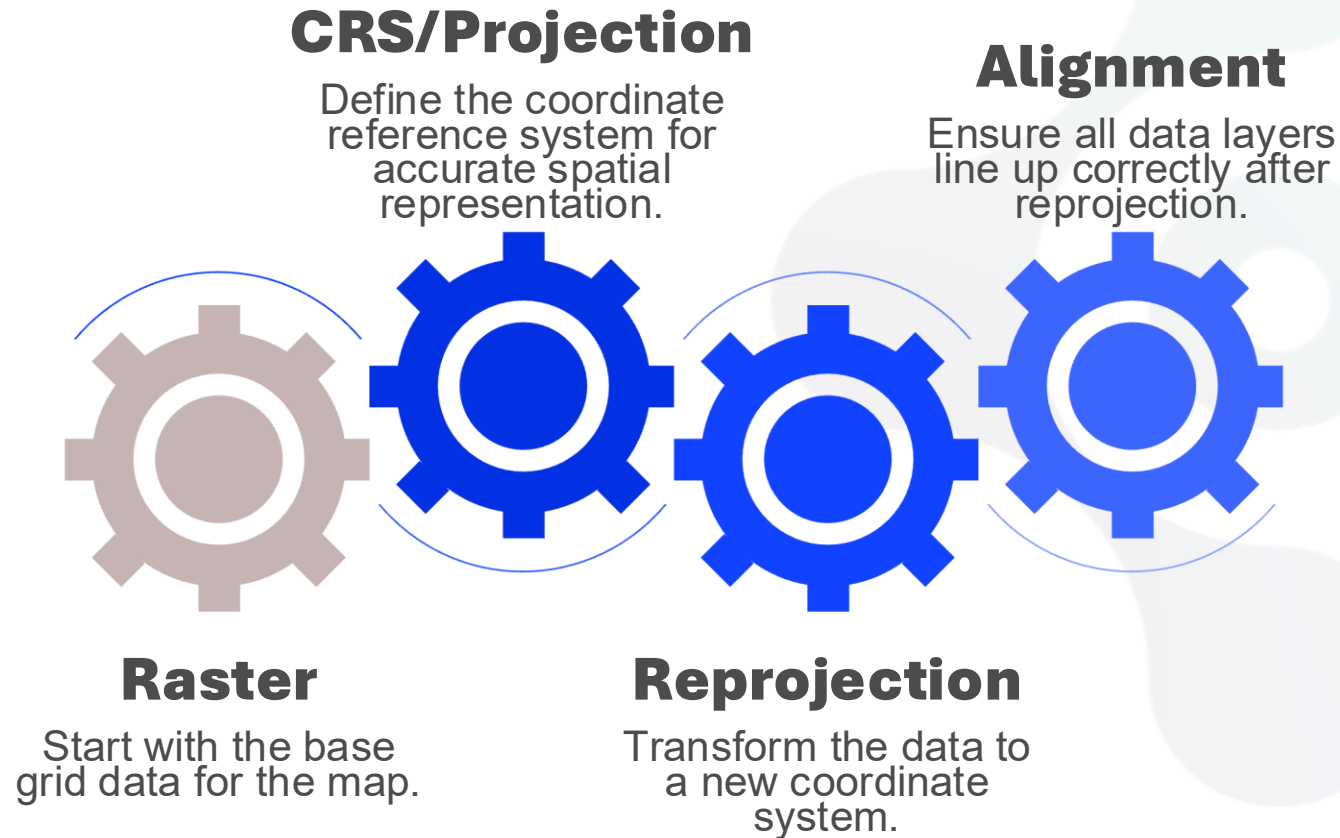
- Multimodal **generative** design
- Shared embedding across modalities
- Thinking-in-Modalities (TiM) capability
- Any-to-any cross-modal generation

 For this workshop, **TerraMind** is the conceptual anchor — its multimodal generative design matches the S1 + S2 + DEM notebook setup directly.



The Geospatial Concepts Hidden Inside the Notebooks

The notebooks assume these concepts. You don't need to master them now — but recognizing them will make the workflow legible.



Raster vs. Vector

Raster = pixel grid (imagery, DEM, masks).
Vector = geometry objects like flood polygons.

Reprojection

Transforming rasters from one CRS to another so S1, S2, and DEM share the same cell grid.

NoData and Masks

Not every pixel is valid. Invalid regions must be tracked — the model cannot use them blindly.

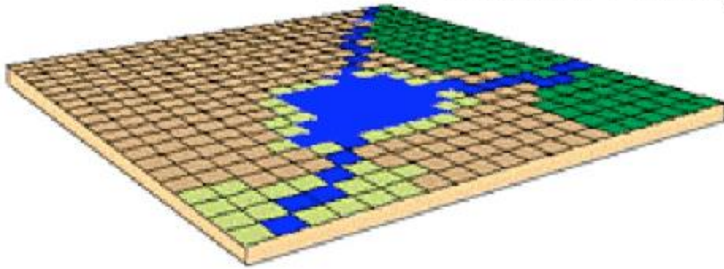
Patch / Chip Extraction

Models consume fixed-size spatial windows. Chips are the unit of inference.

The Geospatial Concepts Hidden Inside the Notebooks

Raster vs. Vector

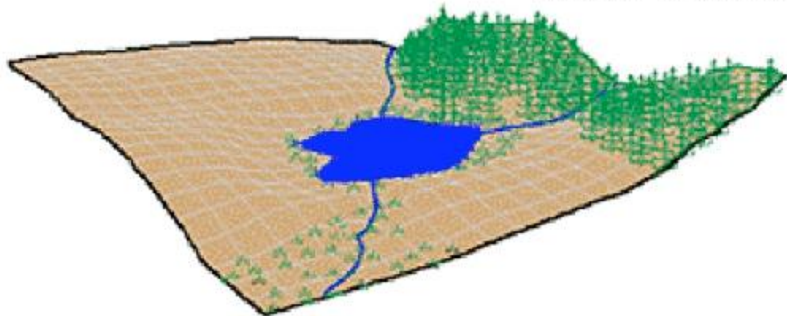
Raster / Image



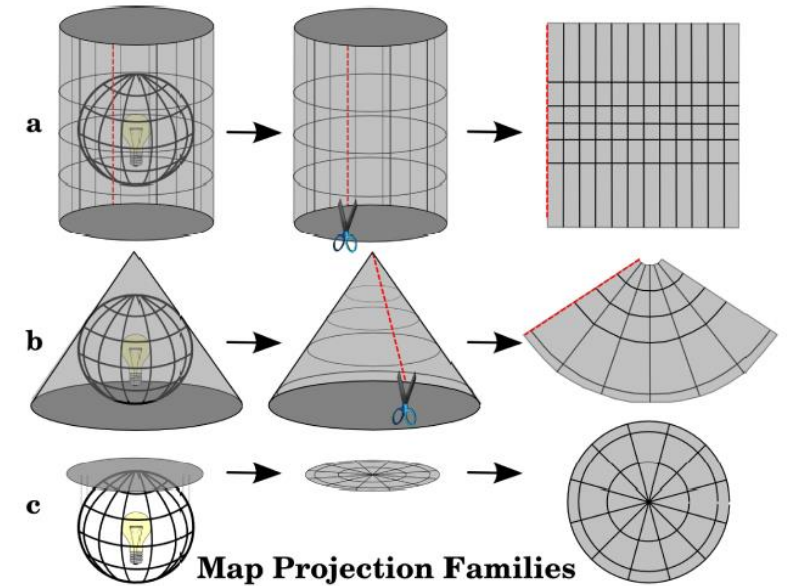
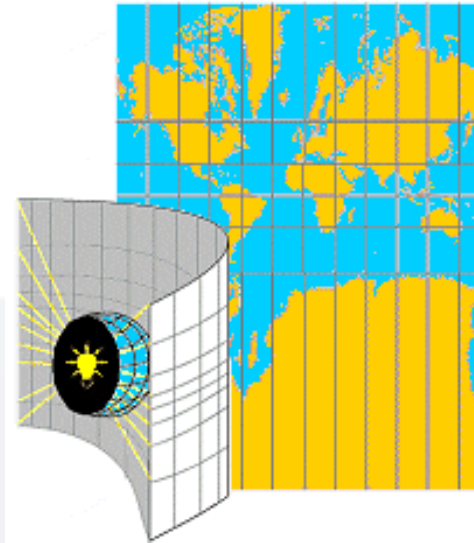
Vector



Real World



(Re)Projection



Map Projection Families



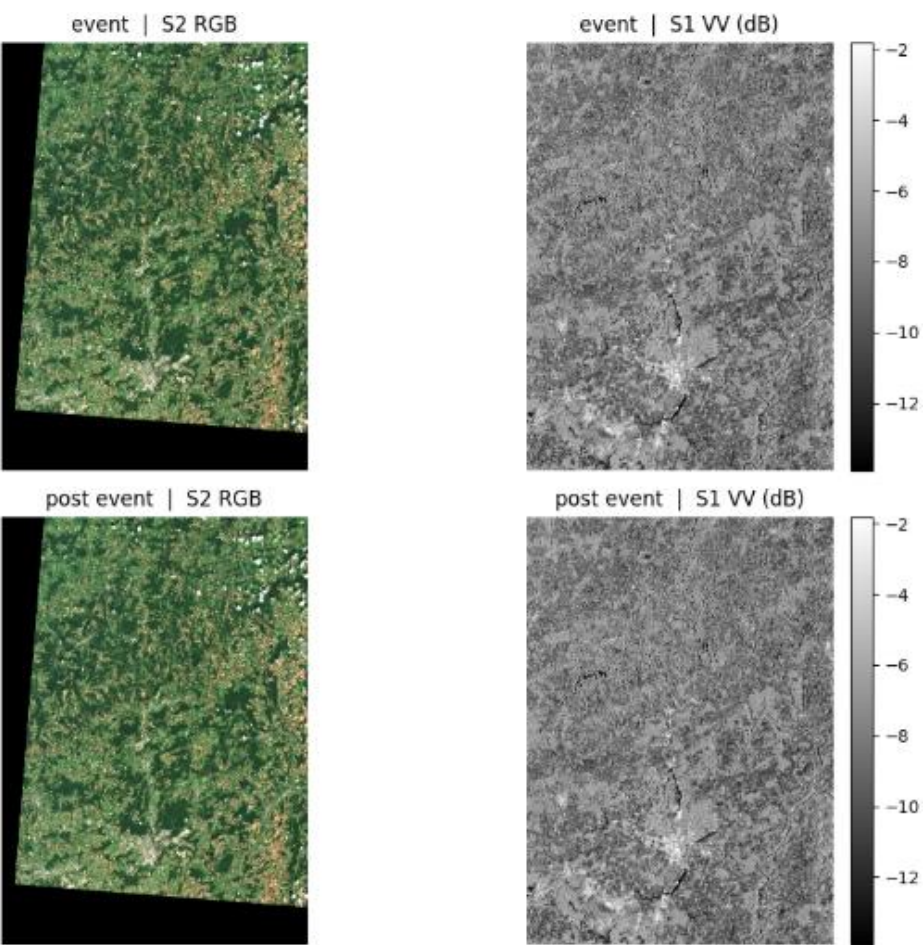
Geographic vs. Projected
Coordinate Reference System (CRS)



The Geospatial Concepts Hidden Inside the Notebooks

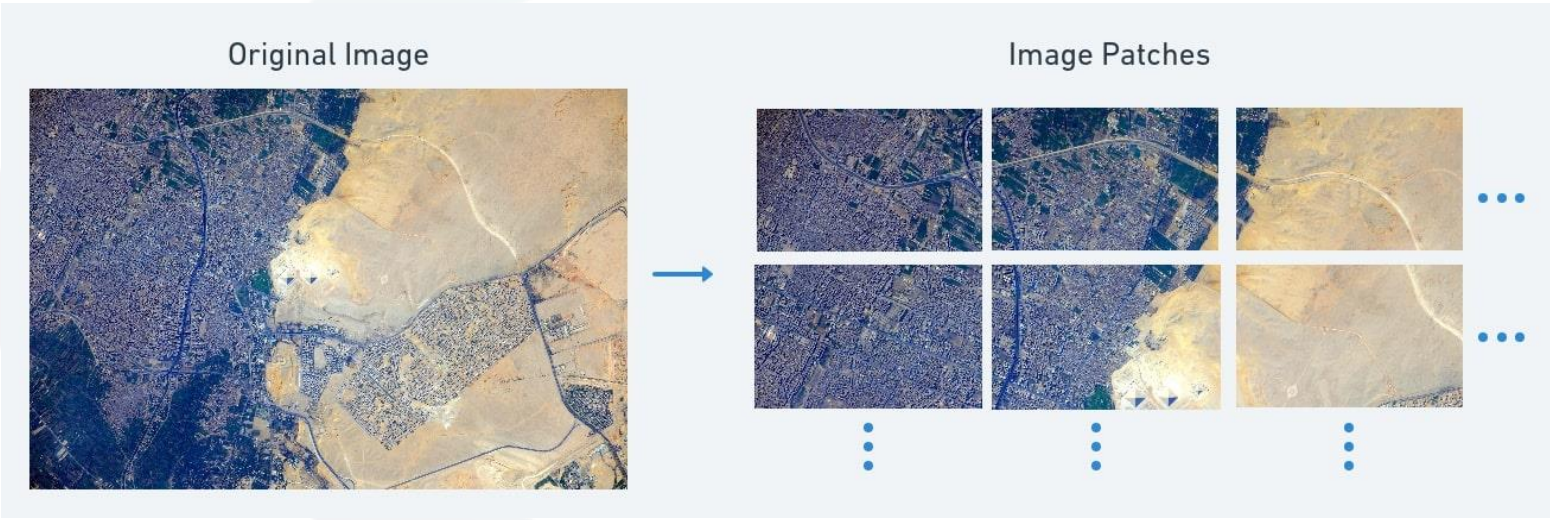
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Patch / Chip Extraction

Models consume fixed-size spatial windows.
Chips are the unit of inference.



256 x 256 or 512 x 512 pixels

What the Three Notebooks Are Really Doing

Each notebook has a specific role in a reproducible, end-to-end geospatial inference pipeline.

Acquire

Collect STAC scenes and Planetary Computer data.

Package

Harmonize layers onto a single analysis grid.

Extract

Crop and validate spatial chips for modeling.

Inference

Classify flood pixels based on multimodal input data





How This Connects to Fine-Tuning on GPUs

LuxProvide Notebooks [01-03]

Data preparation and inference pipeline — the deployment-side of the ecosystem.

- STAC acquisition
- Geospatial harmonization
- Multimodal TerraMind inference

IBM Tutorial Notebook [04]

Training workflow using TerraTorch + PyTorch Lightning.

- Multimodal datamodule
- TerraMind backbone + segmentation decoder
- Lightning trainer with `strategy`, `devices`, `num_nodes`
- Mixed precision for GPU efficiency

 On MeluXina, scaling is controlled by **trainer configuration** — not by rewriting the model. Mixed precision reduces memory footprint and improves throughput.



Hands-On Roadmap After the Concept Section

You don't need to memorize every concept — just recognize its role when it appears in the notebooks.

01

Launch JupyterLab

Open OnDemand on MeluXina — configure and start an interactive GPU job

02

Notebook 1 — Acquire

Review the acquisition manifest, temporal phase logic, and scene ranking

03

Notebook 2 — Prepare

Inspect reprojection, grid alignment, chip extraction, and the valid mask

04

Notebook 3 — Infer

Run TerraMind multimodal flood inference on selected chips

05

Interpret Results

Visualize predictions against input imagery and discuss model limitations

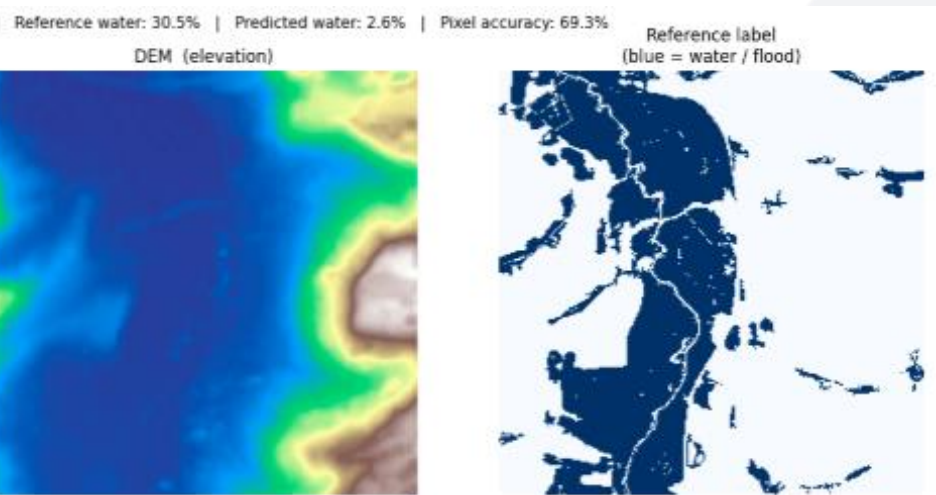


Datasets and Labels in This Tutorial Ecosystem

LuxProvide NOTEBOOKS

Real-World Pipeline Data

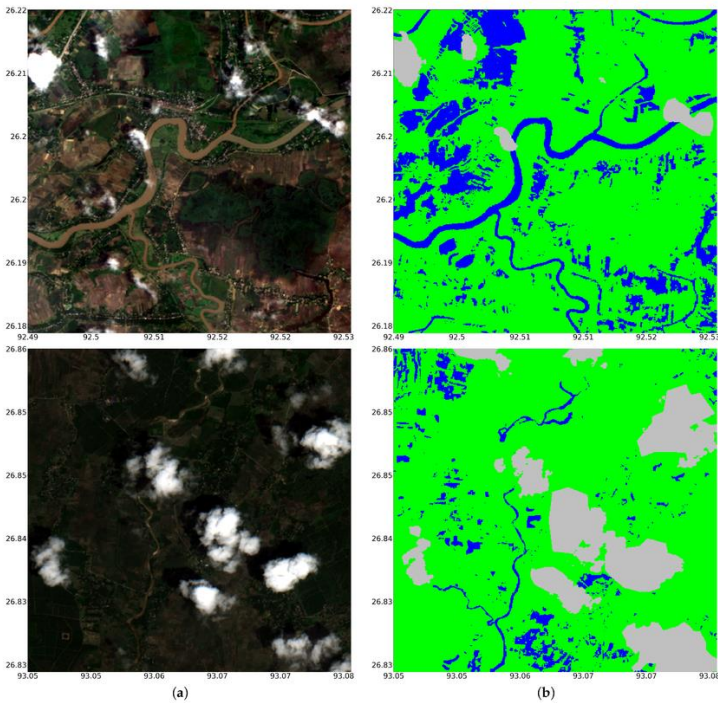
Sentinel-1 RTC, Sentinel-2 L2A, Copernicus DEM, and optional flood delineation reference layers — sourced live via STAC.



IBM TUTORIAL

Sen1Floods11

Flood mapping benchmark: **4,831 chips** at 512×512, spanning 11 flood events across 6 continents. Sentinel-1 centered. Used for fine-tuning examples.



Key distinction: The IBM tutorial is a **fine-tuning example**. The LuxProvide notebooks are a **real data preparation and inference workflow**. Different structures — same model family.

TerraMind for Multimodal Flood Mapping : Luxembourg, July 2021





MeluXina - Open OnDemand (OOD)

Live Open OnDemand walkthrough

<https://portal.lxp.lu>

1. Open OnDemand dashboard → Interactive Apps → JupyterLab
2. Fill in the form live: 1 GPU, 1h 15m walltime, Python venv path
3. Click Launch, then it shows the "queued → starting → running" state transition.

<https://luxprovide.github.io/SCynergy2026-GeoAI/>



Open OnDemand (OOD) Setup - Login

LUXPROVIDE AUTHENTICATION

Sign in to your account

Username or email

Password

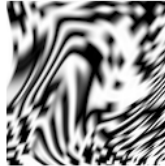
[Forgot Password?](#)

Sign In

KEYCLOAK

Mobile Authenticator Setup

You need to set up Mobile Authenticator to activate your account.

1. Install one of the following applications on your mobile:
FreeOTP
Google Authenticator
2. Open the application and scan the barcode:

[Unable to scan?](#)
3. Enter the one-time code provided by the application and click Submit to finish the setup.
Provide a Device Name to help you manage your OTP devices.

One-time code *

Device Name

Submit

KEYCLOAK

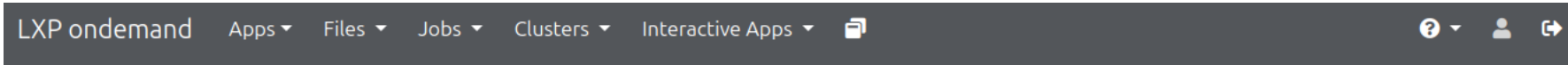
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One-time code

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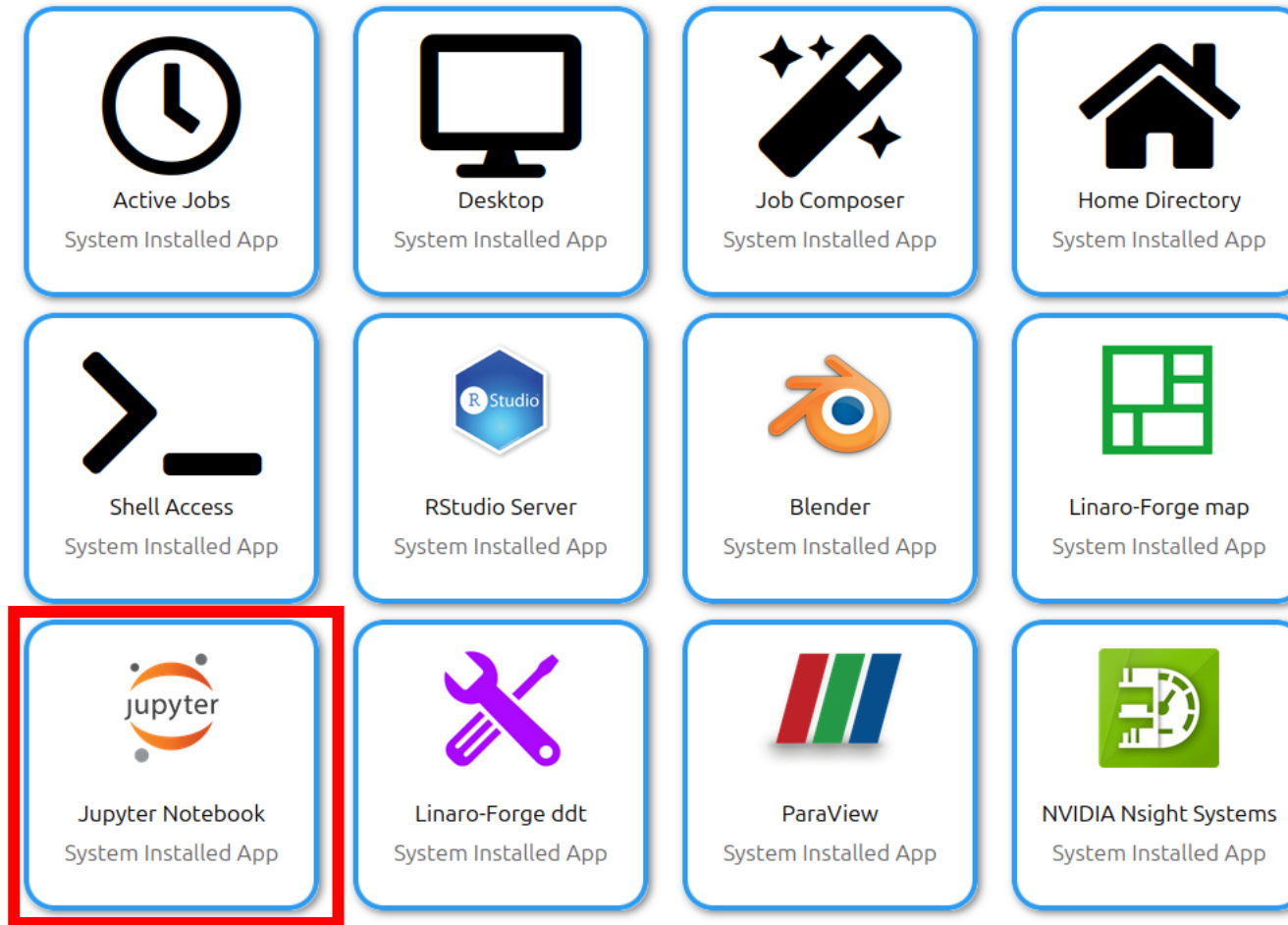


Open OnDemand (OOD) Setup - Apps



OnDemand provides an integrated, single access point for all of your HPC resources.

Pinned Apps A featured subset of [all available apps](#)





OOD JupyterLab – Job Submission Form

Jupyter Notebook

This app will launch a Jupyter Notebook server on one or more nodes.

Account

Partition

Number of Nodes

QOS

Reservation

Timelimit

Software Stack

This defines the default version of Python you want to load. If you want to install different versions of Python, you could install it in your virtual environment.

Built-in Modules to Load

- Account: **[Project ID]**
- Partition: **gpu**
- Number of Nodes: **1**
- QOS: **default**
- Timelimit: **1:30:00**
- Software Stack: **env/release/2025.1**
- Configure Python Environment: **✓**
- MeluXina Preset Python Environment: **SCynergy 2026 - GeoAI**
- Your Python Virtual Environment Path: **e.g., /project/home/[Project ID]/venvs/geoAI**



Takeaways

- **Multimodal** is essential for Floods – We combined **Sentinel-1 (SAR)**, **Sentinel-2 (Optical)**, and **DEM**.
- **TerraMind** is a Geospatial Foundation Model designed to reason across different “modalities” of Earth data.
- **Alignment is Everything**: Earth AI performance starts with geospatial engineering.
- **MeluXina** provides scalable geospatial analytics and has advanced support.

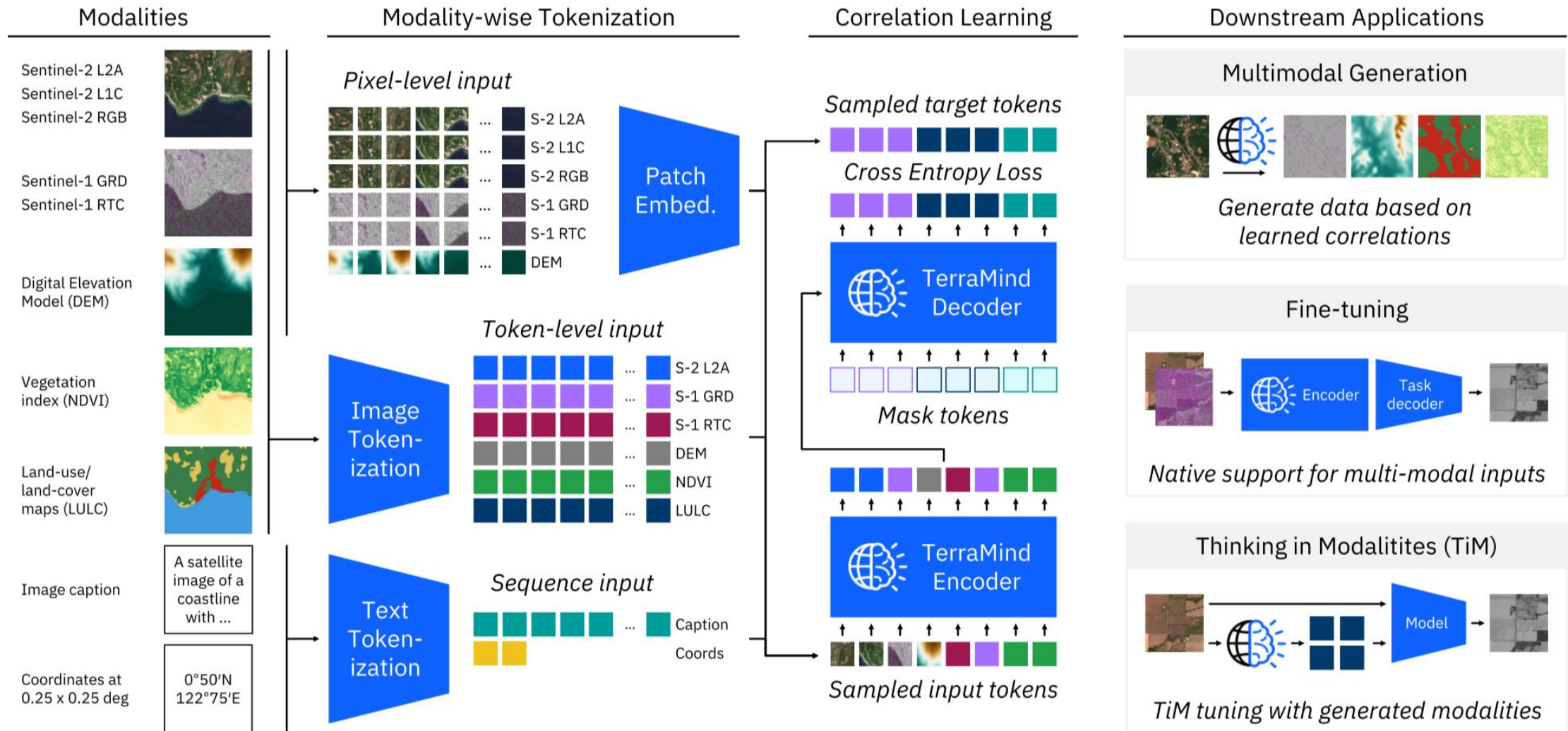


Useful Resources

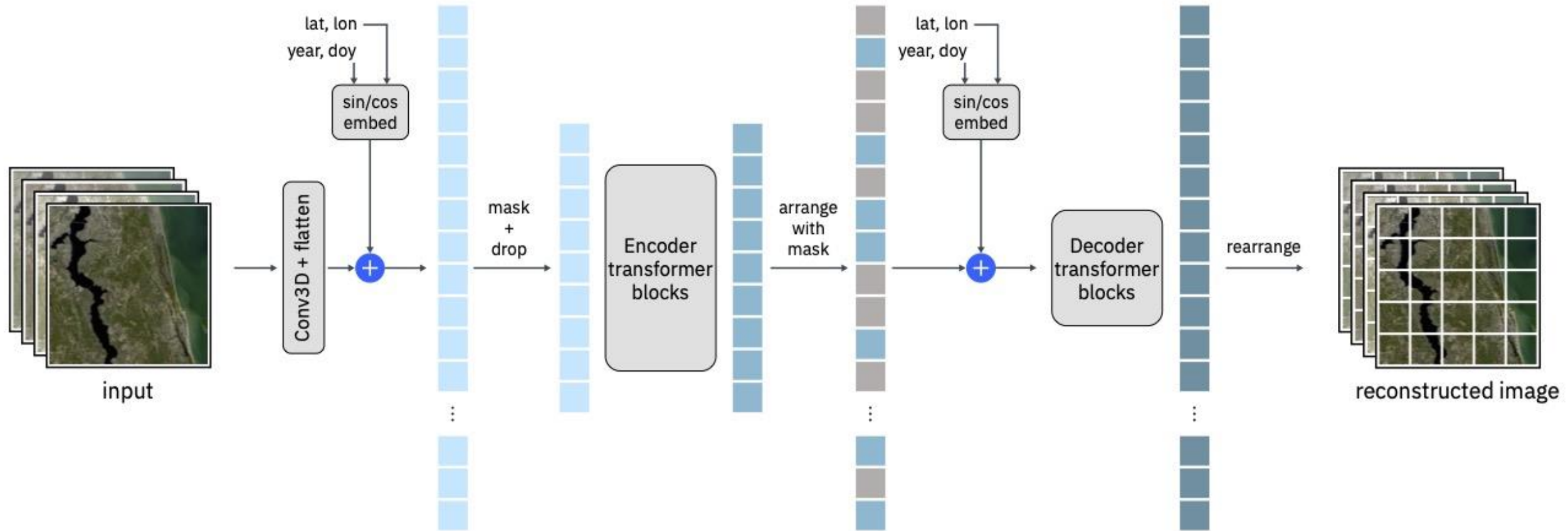
- TerraMind overview: <https://ibm.github.io/terramind/>
- TerraTorch TerraMind guide: <https://terrastackai.github.io/terratorch/1.2.6/guide/terramind/>
- STAC / Planetary Computer tutorial: <https://stacspec.org/en/tutorials/reading-stac-planetary-computer/>
- Raster reprojection with Rasterio: <https://rasterio.readthedocs.io/en/stable/topics/reproject.html>
- Sen1Floods11 benchmark description: <https://experts.arizona.edu/en/publications/sen1floods11-a-georeferenced-dataset-to-train-and-test-deep-learn/>
- MeluXina Open OnDemand JupyterLab docs: https://docs.lxp.lu/web_services/open_ondemand/howtojlab/
- MeluXina Open OnDemand JupyterLab – environment setup: https://docs.lxp.lu/web_services/open_ondemand/jlab_env/



Supplementary: TerraMind



Supplementary: Prithvi-EO-v2





SCynergy 2026



EPICURE

Unlocking European-level HPC Support

EXPLORING THE FRONTIERS OF DIGITAL INTELLIGENCE

